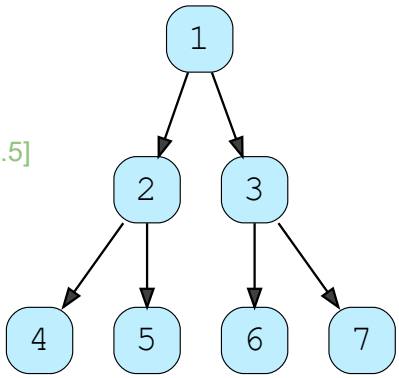
Solution: Level Averages in a Binary Tree

Problem Statement

Given a binary tree, populate an array to represent the **averages of all of its levels.**

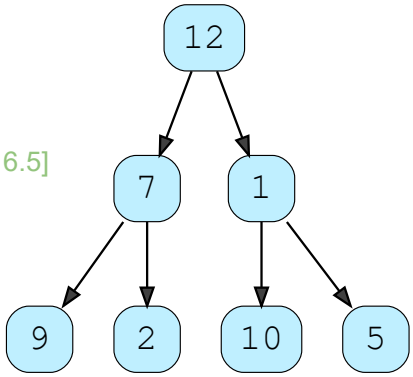
Example 1:

Level Averages: [1, 2.5, 5.5]



Example 2:

Level Averages: [12.0, 4.0, 6.5]

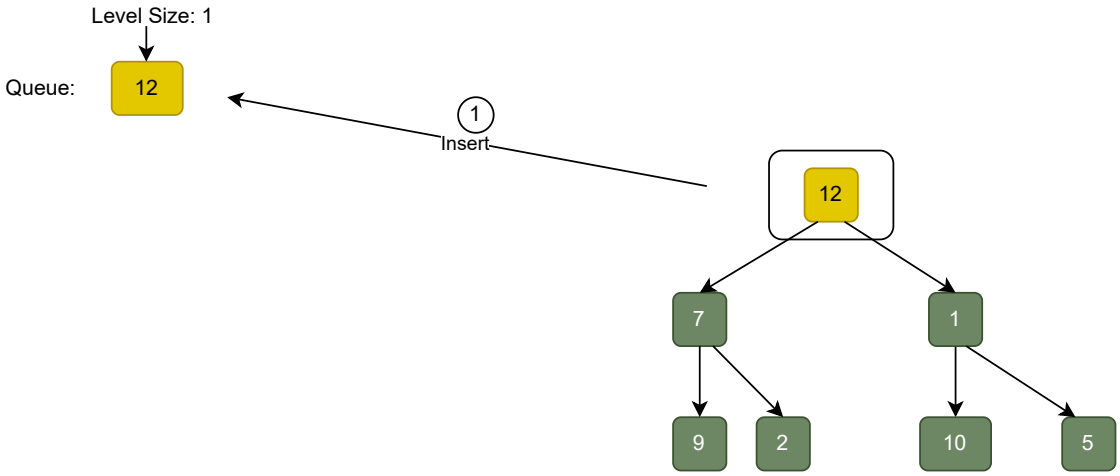


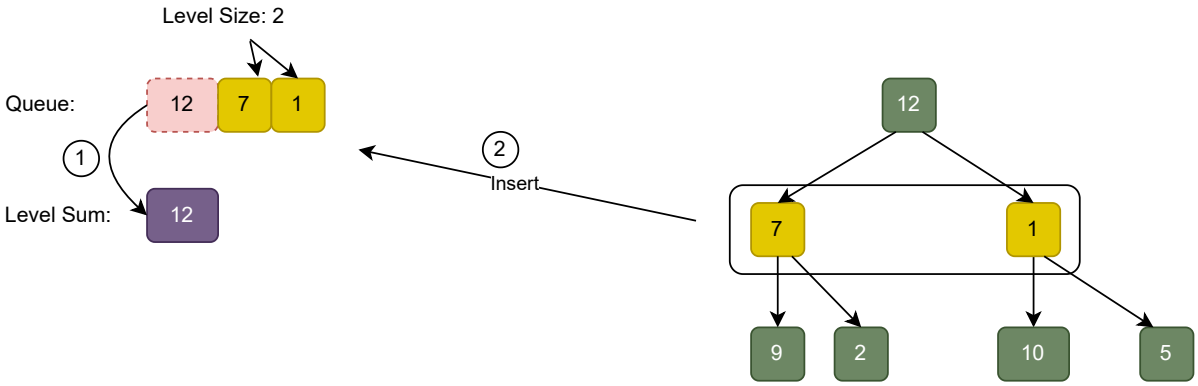
Solution

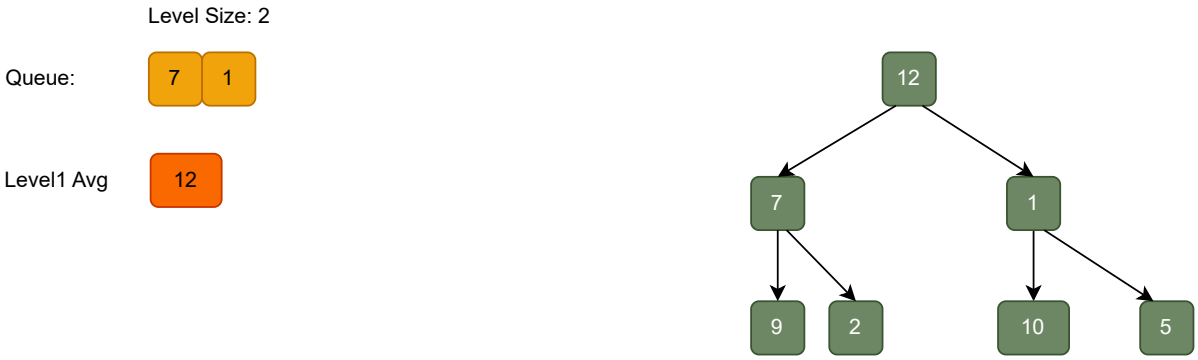
This problem follows the **Binary Tree Level Order Traversal** pattern. We can follow the same **BFS** approach. The only difference will be that instead of keeping track of all nodes of a level, we will only

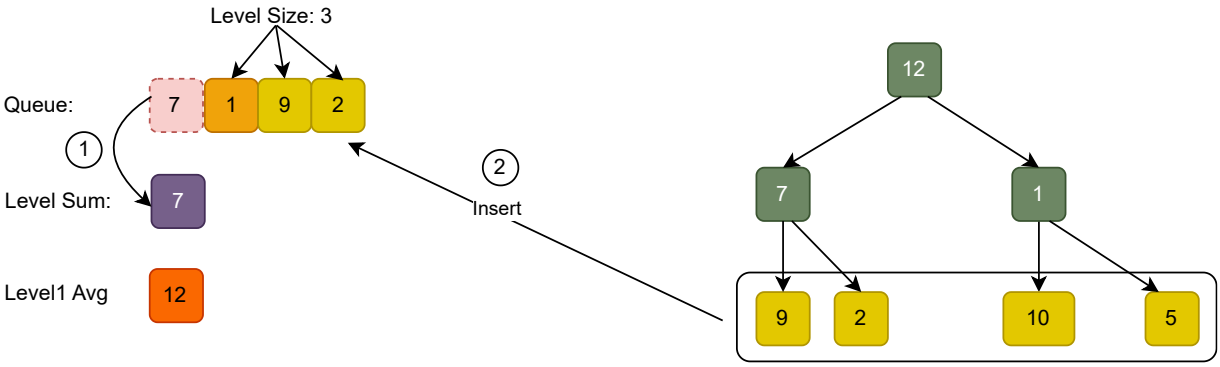
track the running sum of the values of all nodes in each level. In the end, we will append the average of the current level to the result array.

Here is the visual representation of the algorithm:

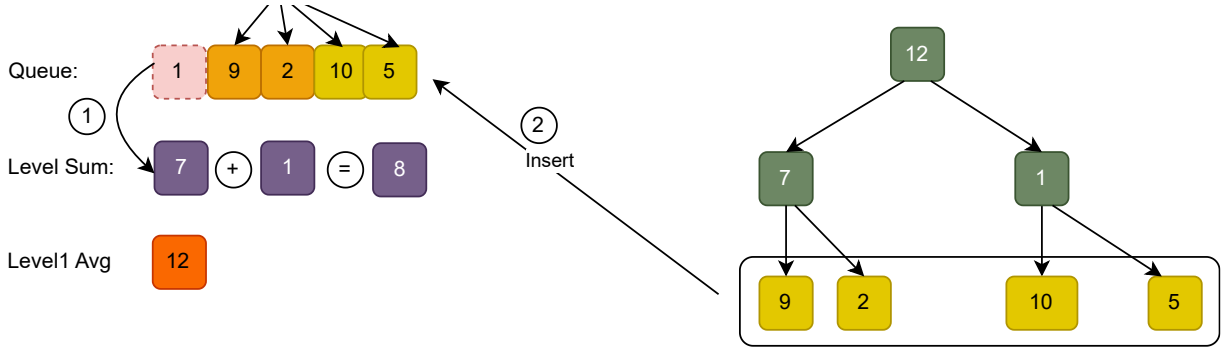

Level Size: 1
Queue:
12
1
Insert
12
7
1
9
2
10
5

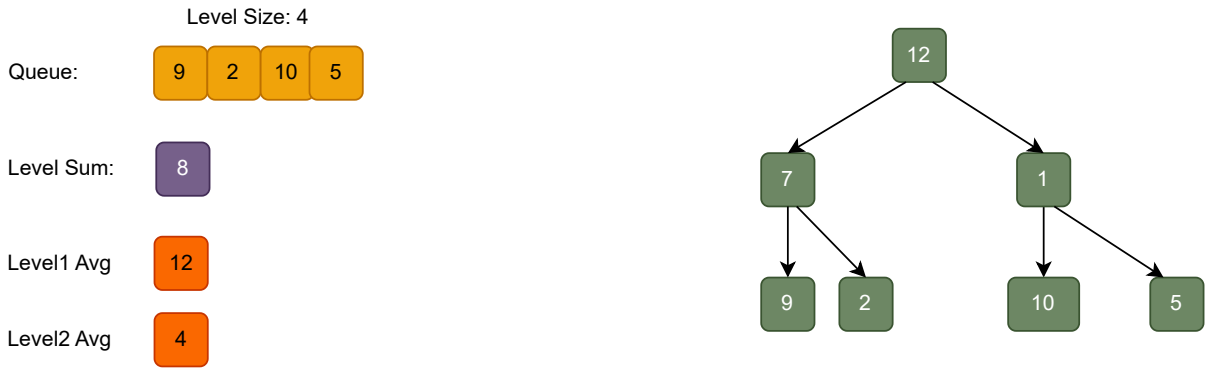

Level Size: 2
Queue:
12
7
1
1
Level Sum:
12
2
Insert
12
7
1
9
2
10
5

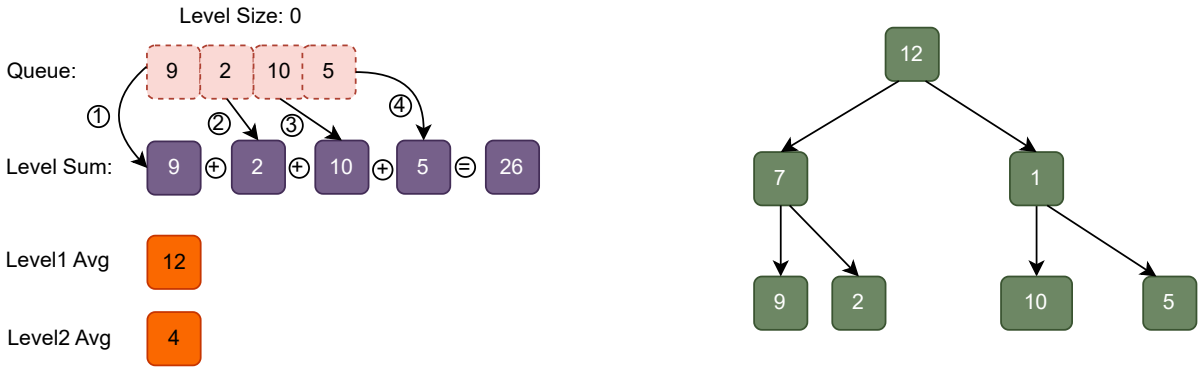

Level Size: 2
Queue:
7
1
Level1 Avg
12
12
7
1
9
2
10
5


Level Size: 3
Queue:
7
1
9
2
1
Level Sum:
7
Level1 Avg
12
2
Insert
12
7
1
9
2
10
5


Level Size: 4





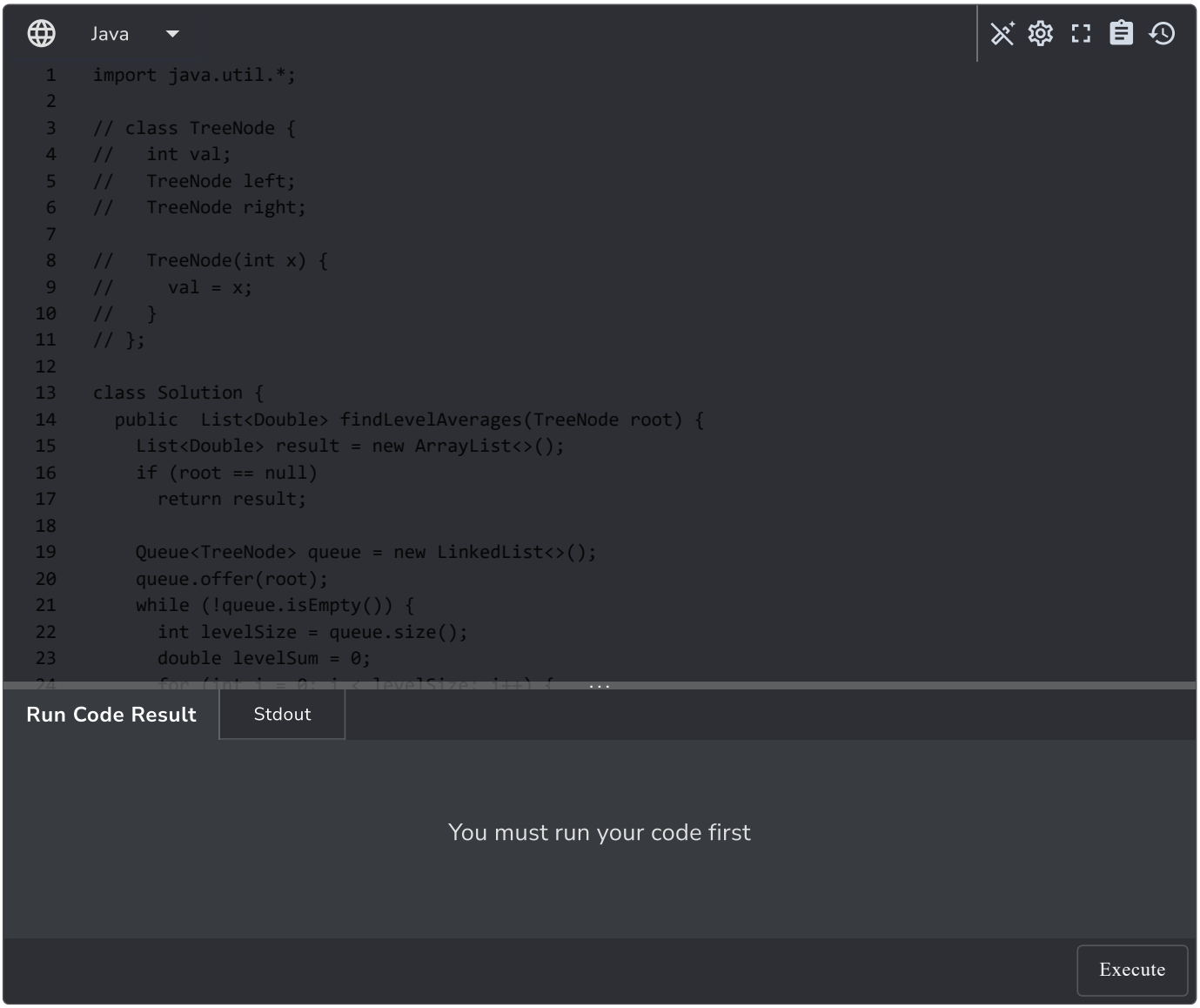




Level Averages in a BT

Code

Here is the code for this algorithm:



Time Complexity

The time complexity of the above algorithm is $O(N)$, where N is the total number of nodes in the tree. This is due to the fact that we traverse each node once.

Space Complexity

The space complexity of the above algorithm will be $O(N)$ which is required for the queue. Since we can have a maximum of $N/2$ nodes at any level (this could happen only at the lowest level), therefore we will need $O(N)$ space to store them in the queue.

Similar Problems

Problem 1: Find the largest value on each level of a binary tree.

Solution: We will follow a similar approach, but instead of having a running sum we will track the maximum value of each level.

```
maxValue = max(maxValue, currentNode.val)
```

← Previous

Level Averages In A Binary Tree (Easy)

Next →

Minimum Depth Of A Binary Tree (Easy)

Mark as Completed